Activity Report

On

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

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LIQUID CRYSTAL DISPLAYS

(Basic Electronics)

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Nov 2023

INTRODUCTION

Imagine a world without vibrant displays on your smartphones, computers, and televisions. That's the world we would have lived in if not for Liquid Crystal Displays (LCDs), the unsung heroes of the modern digital age.

LCDs are flat-panel display technologies that utilize the unique properties of liquid crystals, a special class of materials that exhibit both fluid and solid-like behaviors. By manipulating the alignment of liquid crystal molecules using electric fields, LCDs can selectively control the passage of light, creating visible images.

Picture a sandwich made up of two polarizing filters, a layer of liquid crystals, and a backlight. When light passes through the first polarizer, it becomes polarized, meaning its vibrations are aligned in a specific direction. As the light enters the liquid crystal layer, its molecules align themselves according to the applied electric field. If the molecules are aligned in a way that allows the polarized light to pass through, it reaches the second polarizer and exits, producing a visible pixel on the screen. However, if the liquid crystals are not aligned, the light is blocked, resulting in a dark pixel.

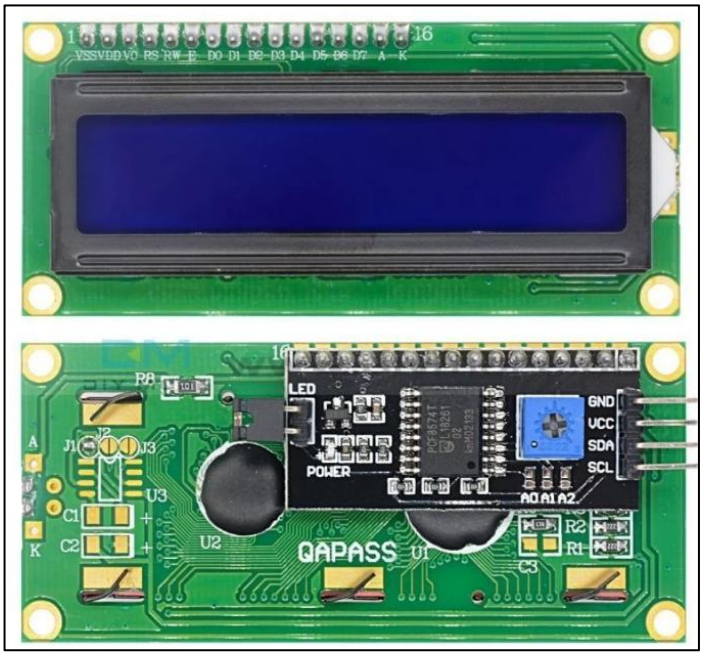

Fig. 1.1 LIQUID CRYSTAL DISPLAY

THEORY

A liquid-crystal display (LCD) is a flat-panel display or other electronically modulated optical device that uses the light-modulating properties of liquid crystals combined with polarizers. LCDs can either be normally on (positive) or off (negative), depending on the polarizer arrangement. For example, a character positive LCD with a backlight will have black lettering on a background that is the color of the backlight, and a character negative LCD will have a black background with the letters being of the same color as the backlight.

Principle of LCD:

The principle behind the LCDs is that for switching pixels on and off to reveal a specific color, LCD screens use liquid crystals. And, the molecule tends to untwist at the time of electrical current is applied to the liquid crystal molecule. This becomes the reason for a change in the angle of the top polarizing filter as well causes the angle of light, a light that is passing through the molecule of the polarized glass. Consequently, with the help of an individual area of the LCD, a small light is allowed to permit the polarized glass.

Construction:

A reflected mirror is arranged at the backside at the time of constructing the LCDs. An indium-tin-oxide is used to make the electrode plane, which is kept on top of the device. Also, on the bottom of the device, a polarized glass with a polarizing film is added. With the help of a common electrode, the LCD's complete area has to be enclosed, and the liquid crystal matter should be above it.

Then comes another polarizing film on top, and on the bottom, in the form of rectangle, the second piece of glass comes with an electrode. But make sure that both pieces are located on the right angles. The light passes through the front of the LCD when there is no current, reflecting with the help of a mirror and bounced back. A screen is in front of the light that is made up of pixels colored red, green, and blue. In order to reveal a certain color or keep that pixel black, the liquid crystals work for turning a filter on or off.

CIRCUIT DIAGRAM

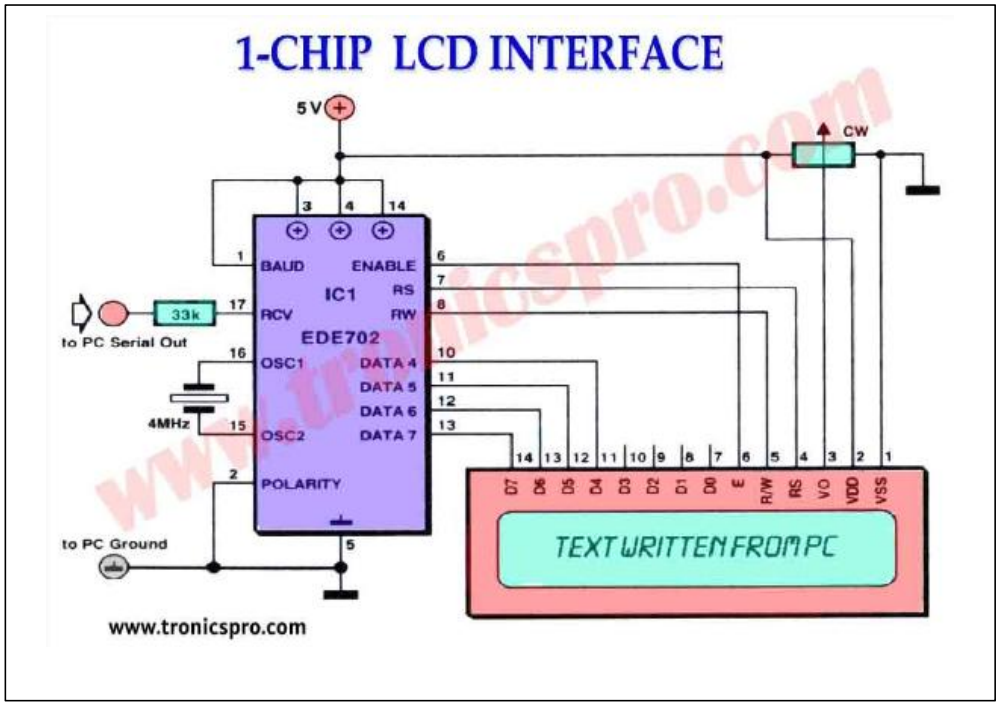

Fig.1.2 CIRCUIT DIAGRAM

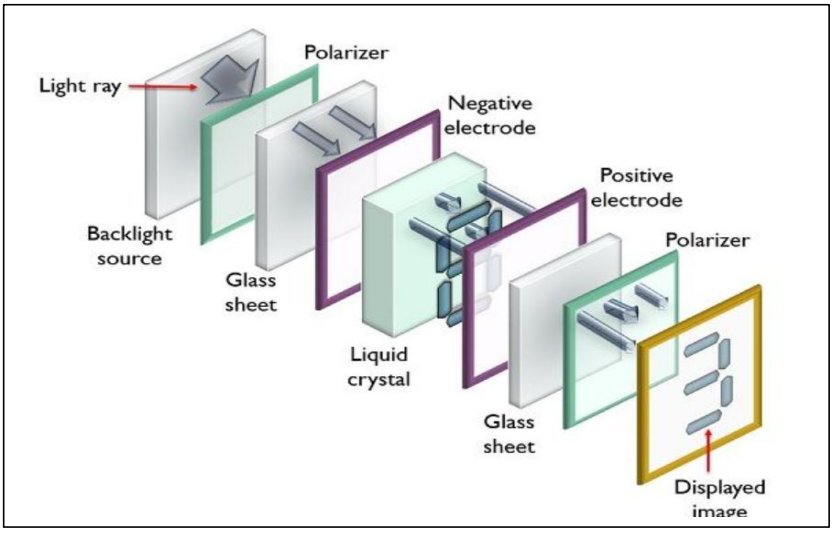

Fig. 1.3 CONSTRUCTION

Construction of LCD consists of two polarized glass pieces. Two electrodes are used, one is positive and the other one is negative. External potential is applied to LCD through this electrodes and it is made up of indium-tin-oxide. Liquid crystal layer of about 10µm- 20µm is placed between two glass sheets. The light is passed or blocked by changing the polarization.

OPERATION PRINCIPLE

Liquid Crystal display, is a technology used in diverse digital devices like flat-panel TVs, PC monitors, smartphones, and more. LCD s work based totally on the principle of manipulating the homes of liquid crystals to manipulate the passage of mild and create pictures. here's how they operate and their primary concepts:

Liquid Crystals: LCD s use a cloth known as liquid crystals. These are materials which have homes of each liquid and solids. when an electric subject is applied, the molecules of the liquid crystals align in a selected course.

Polarization: In a liquid crystal display, there are layers of polarized glass on either facet of a layer of liquid crystals. Polarization filters are used to control the path of mild passing via the LCD. One layer has vertical polarization, and the alternative has horizontal polarization.

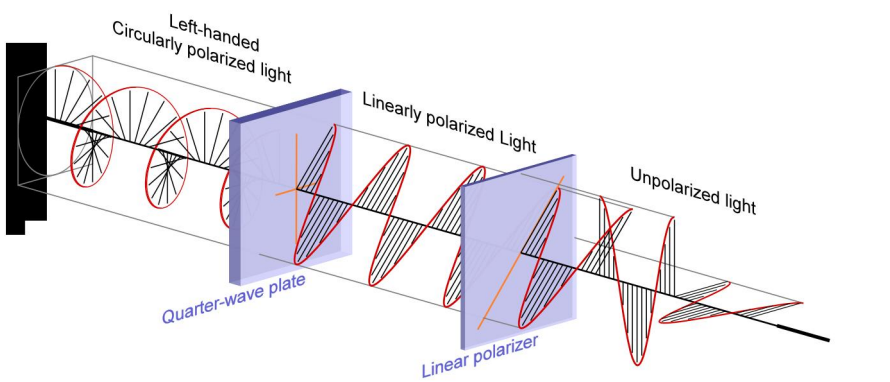

Fig. 1.4 POLARIZING FILTER

Liquid Crystal Alignment: The liquid crystal layer sits among these polarized layers. When no voltage is implemented, the liquid crystal molecules are aligned in a manner that does not disturb the polarization of light passing via.

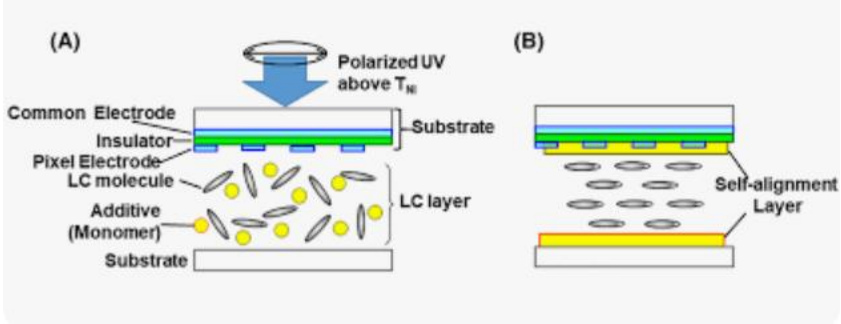

Fig. 1.5 ALIGNMENT

Voltage Manipulate: To create photographs or manipulate the amount of mild passing via the liquid crystal display, a voltage is carried out to particular segments of the liquid crystal layer. This voltage adjusts the orientation of the liquid crystal molecules, which, in flip, alters the polarization of light passing via that section.

Photo Introduction: by selectively making use of voltage to exceptional segments of the liquid crystal layer, the LCD can manipulate the amount of mild passing via every segment, allowing the introduction of photographs with distinctive hues and brightness tiers.

Shade Filters: most liquid crystal display presentations use colour filters to produce a full range of colors. Those filters are commonly purple, green, and blue (RGB). by means of various the intensity of light passing through every coloration clear out, a complete spectrum of colors can be displayed.

Backlight: In most LCD s, there may be a backlight supply behind the liquid crystal layer. This offers the illumination for the display. The liquid crystals control how a whole lot of this light is allowed to bypass thru and create the picture.

Pixels: Liquid crystal display presentations are composed of many character pixels (picture factors), every of which includes 3 sub-pixels (red, inexperienced, and blue). With the aid of controlling the nation of the liquid crystals in every sub-pixel, the liquid crystal display can generate extraordinary shades and sun shades.

The aggregate of specific voltage manage, liquid crystal orientation, and colour filtering allows LCD s to create high quality photographs and videos on a spread of monitors. Liquid crystal display technology is known for its electricity performance, quite low cost, and its capacity to supply bright and sharp snapshots, making it broadly used in a variety of purchaser electronics and shows.

APPLICATIONS

Some daily life applications of LCD include:

- Clocks: Used in displays in digital wall clocks
- Cameras: Used in digital cameras for displaying images
- Television: Has a major application in LCD TFT screens
- Computer monitors: Most of the modern world computer monitors are made of LCD s
- Instrument panels: All the lab instruments uses LCD screens for display
- Aircraft cockpit displays: Cockpit display panels of pilots are usually LCD s
- Video players: Such screens are very popular in video game parlour and consoles
- Watches: All the digital wrist watches uses LCD screen for displaying time
- Calculators: Calculator display screens are LCD screens

Fig.1.6 APPLICATIONS OF LCD

CONCLUSION

LCDs have become the dominant display technology due to their numerous advantages over traditional cathode-ray tube (CRT) displays:

Compact Size and Lightweight: LCDs are remarkably thin and lightweight, making them ideal for portable devices and space-constrained environments.

Low Power Consumption: LCDs consume significantly less power than CRTs, contributing to longer battery life and lower energy bills.

Superior Image Quality: LCDs offer high resolution, wide viewing angles, and consistent color reproduction, resulting in superior image quality.

Reduced Environmental Impact: LCDs contain fewer hazardous materials and require less manufacturing energy compared to CRTs.

LCD technology is constantly evolving, with researchers exploring new materials, fabrication processes, and display architectures. These advancements aim to:

Enhance image quality with higher resolution, wider color gamuts, and improved contrast ratios.

Reduce power consumption for longer battery life and energy efficiency.

Expand viewing angles for more immersive experiences.

Develop flexible and transparent LCDs for innovative applications.

As LCD technology continues to progress, we can expect even more breathtaking visuals and versatile displays that seamlessly integrate into our lives.

It's evident that LCDs have become indispensable in our daily lives, offering vibrant colors, sharp resolutions, and energy efficiency. The versatility of LCD technology has not only transformed entertainment and communication but has also played a crucial role in fields such as healthcare, education, and industry.

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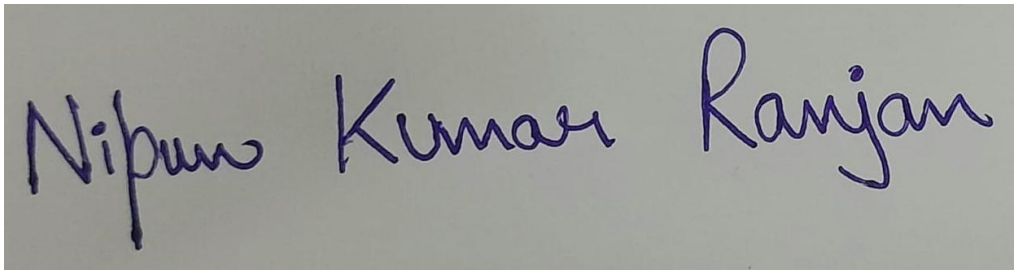
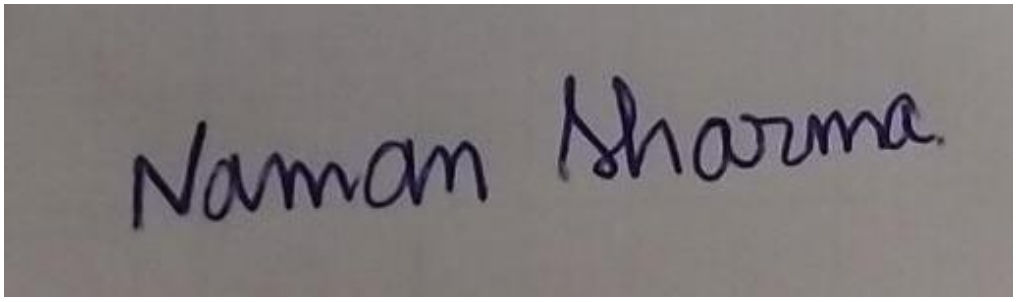
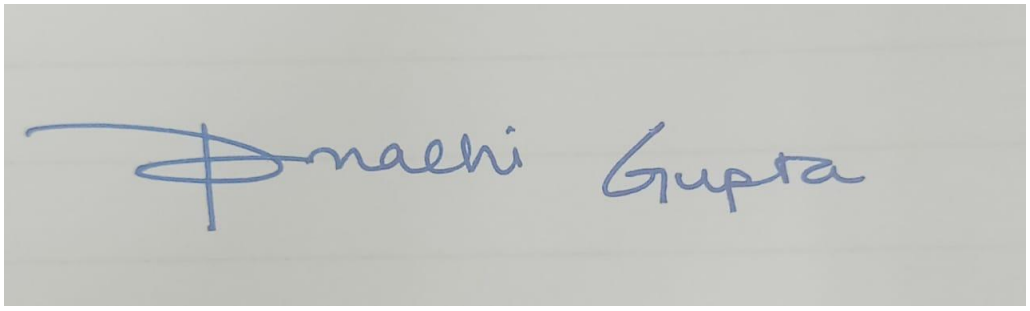
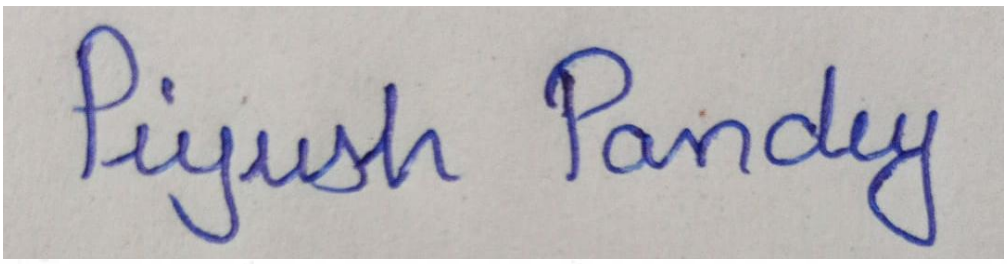
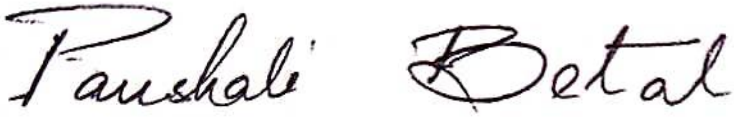
STUDENT SIGNATURES











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